Coastal Hydrodynamics II

Course : CE60222

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Content

- Wave Skewness And Asymmetry
- Wave Orbital Velocity
- Dynamic Pressure
- Wave Boundary Layer
- Bed Shear Stress

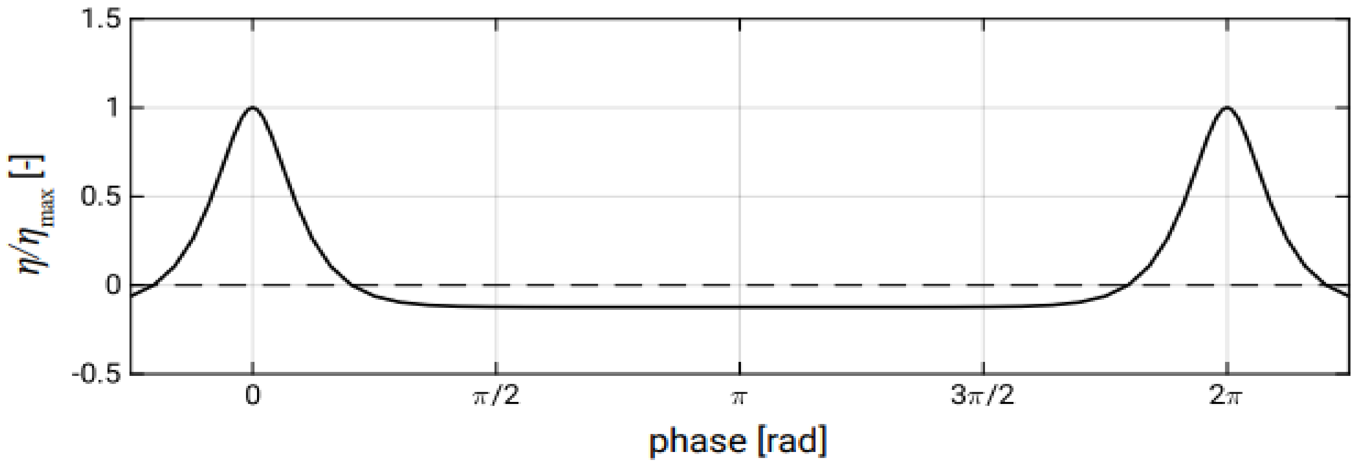
Wave skewness and Asymmetry

Skewness:

- Gradual **peaking of the wave crest** and a **flattening of the trough.** This asymmetry relative to the **horizontal axis** is called **skewness**.

Asymmetry:

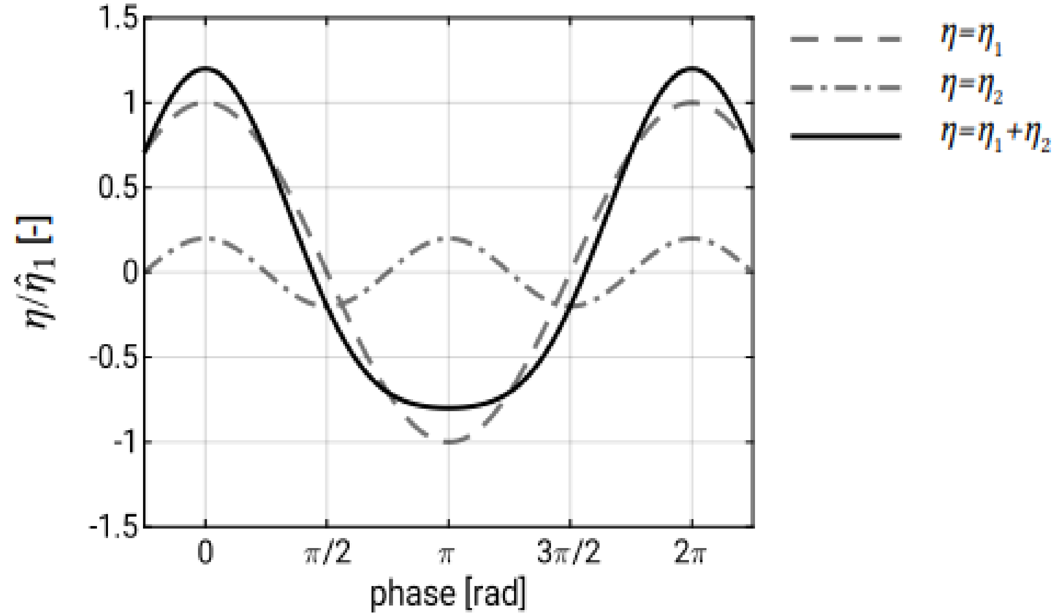
- **Relative steepening** of the face until breaking occurs, resulting in a **pitched-forward wave shape. This asymmetr**y relative to the **vertical axis** is often simply called **asymmetry**.



- Waves propagating towards the shore become more and more asymmetric, until the point of wave-breaking. These non-linear effects cannot be described by linear theory.
- Numerous non-linear theories (Stokes theory, cnoidal wave theory, Boussinesq equations) have been developed to take into account these complicated non-linear processes.
- The non-linear effects are crucial in determining the magnitude of the wave-induced transport. In the following we will therefore subsequently consider skewness and asymmetry.
- This asymmetric profile (relative to the horizontal axis) can only be described by a sum of sinusoidal waves with higher harmonics (frequencies that are a multiple of the basis frequency: $\cos S$, $\cos 2S$ etc. with $S = \omega t - kx$)
- The second-order equation for the surface elevation can be written as:

$$\boldsymbol{\eta = \hat{\eta}_1 cos(\omega t - kx) + \hat{\eta}_2 cos2(\omega t - kx)}$$

- The amplitude of the second-order correction is small compared to the first-order component.
- It can be seen that the resulting Stokes wave profile $\eta_1 + \eta_2$has crests which are narrower and more peaked than those of a cosine profile and troughs that are wider and fatter; the profile is skewed.
- The second term η_2 represents the Stokes second-order wave travelling at the same speed as the first-order wave η_1 (hence, it does not obey the linear dispersion relation).



- The pitched-forward wave shape is the result of the fact that in shallow water the wave crest moves faster than the wave trough.

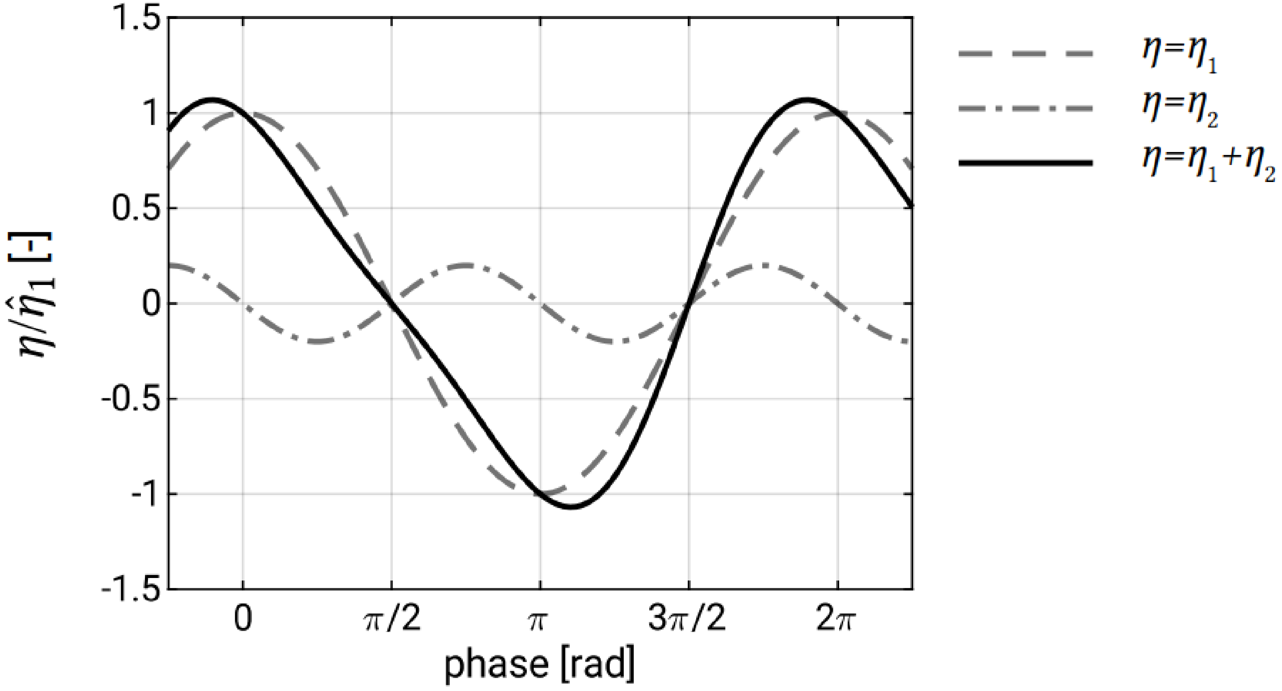
- The propagation speed of non-linear shallow-water waves

$$\boldsymbol{c} = \sqrt{\boldsymbol{g}(\boldsymbol{h} + \boldsymbol{\eta})}$$

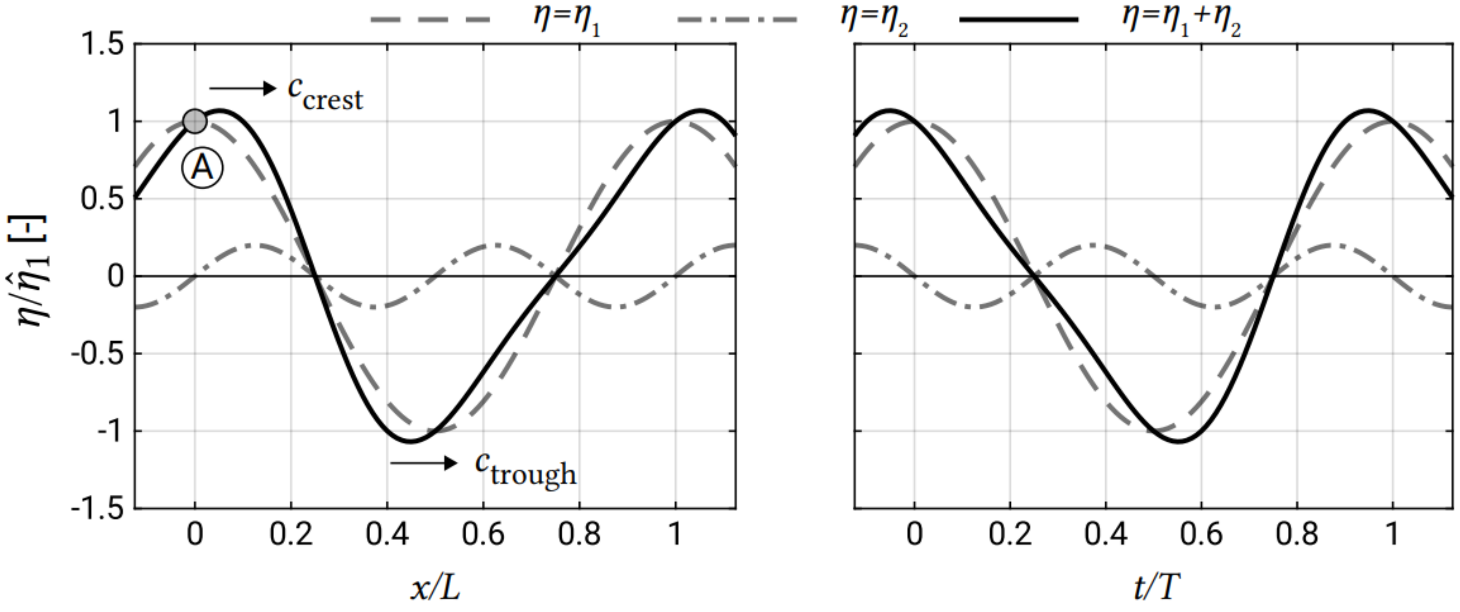
- For small-amplitude shallow-water waves this reduces to $\boldsymbol{c} = \sqrt{\boldsymbol{gh}}$

- The wave crest of a harmonic wave with amplitude a has a higher propagation velocity $c_{crest} = \sqrt{g(h+a)}$ than the trough which propagates with $c_{trough} = \sqrt{g(h-a)}$.

- Closer to the **surf zone**, **phase-shifting** of the **harmonic(s)** leads to an **increase** in **wave asymmetry** and – eventually – to **a decrease in wave skewness** as well.
- Ultimately the **pitching forward** results in **wave-breaking**.

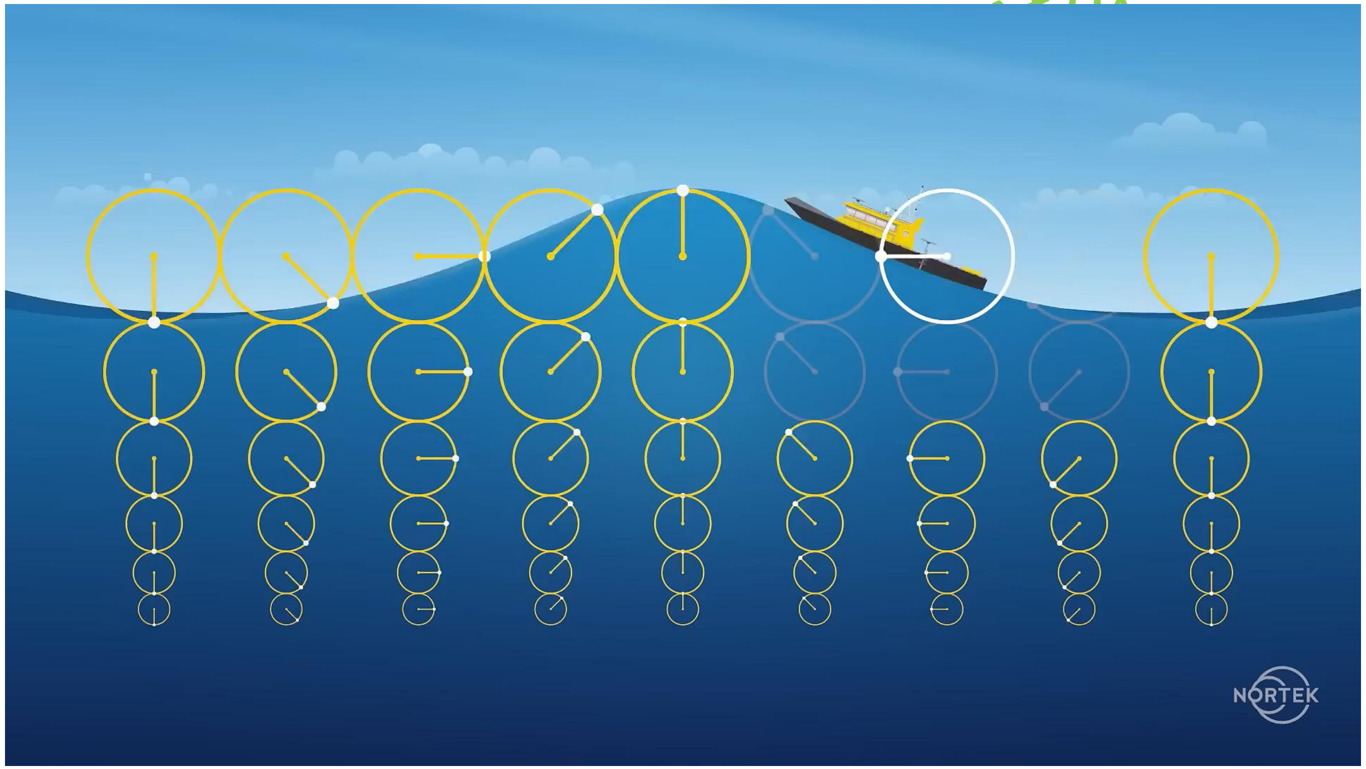


- The right panel of Figure shows a time series of a pitched-forward (sawtooth-like) wave at one location, namely location A ($x/L = 0$) in the left panel of Figure.
- The left panel of Figure shows a rapidly rising and slowly falling surface elevation.
- The sawtooth-like wave of Figure, the phase shift between the first and the second harmonic is such that the asymmetry is maximum and the skewness is zero.
- The sawtooth-like wave of Figure can also be depicted as a function of $S = \omega t - kx$.

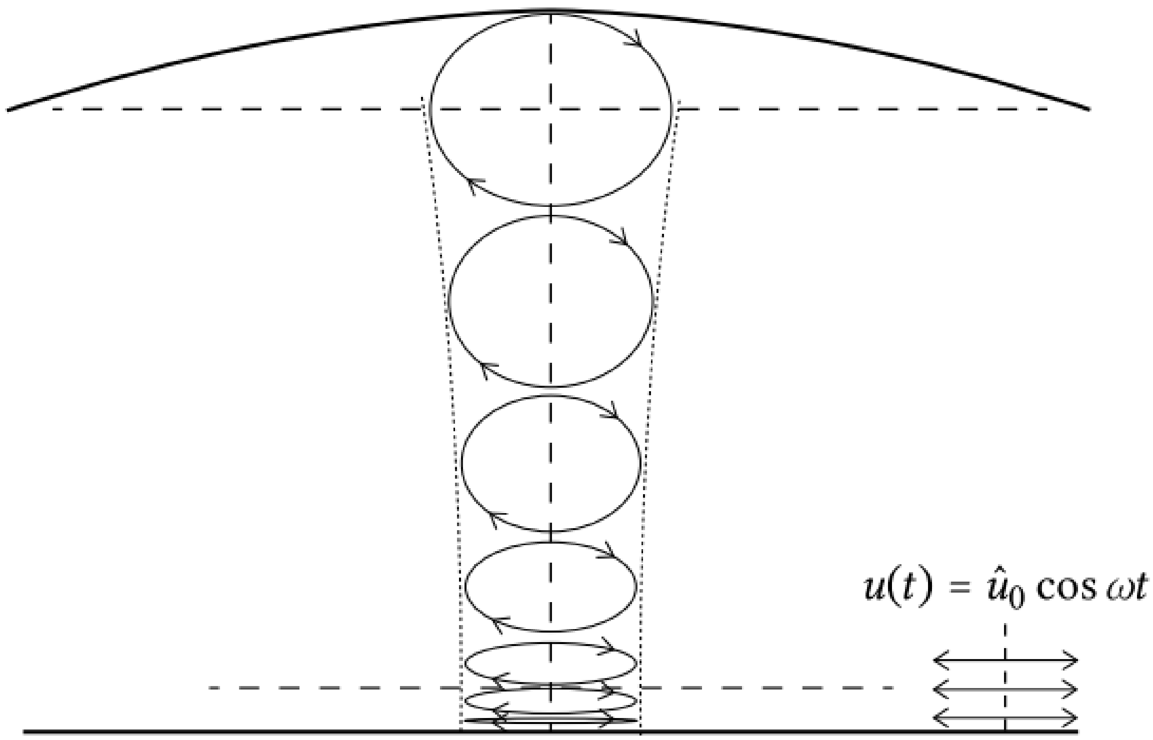


Wave orbital velocity

- Wave orbital velocity refers to the movement of water particles beneath the surface of a wave as it propagates through the ocean.
- As a wave travels, the water particles do not move in a straight line. Instead, they follow a circular or elliptical path. This motion is called "orbital" because the particles move in orbits.



- In deep water, the orbits of the water particles are nearly circular. At a depth of about half a wavelength, the circular motion is reduced to about 4% of the surface value.
- In shallow water, the orbits become elliptical. The water particles still move up and down, but the motion is less circular and more flattened as they approach the seabed. At the bottom, the vertical movement is zero.



- Understanding wave orbital velocity is crucial for coastal engineering, sediment transport studies, and predicting how waves interact with the shoreline.
- The horizontal velocity of the water particles varies with depth and is influenced by the wave's characteristics, such as its amplitude and wavelength.
- The velocity can be described mathematically, showing how it changes with depth.
- Now consider waves of infinitesimal amplitudes. According to linear theory, the horizontal orbital velocity varies harmonically with an amplitude $\hat{u}$ equal to:

$$\hat{\boldsymbol{u}}(\boldsymbol{z}) = \boldsymbol{\omega a} \frac{\boldsymbol{\cosh k(h+z)}}{\boldsymbol{\sinh kh}}$$

where: $\boldsymbol{\omega}$ = angular frequency ($2\pi/T$) rad/s

$\boldsymbol{a}$ = wave amplitude m

$\boldsymbol{k}$ = wavenumber ($2\pi/L$) rad/m

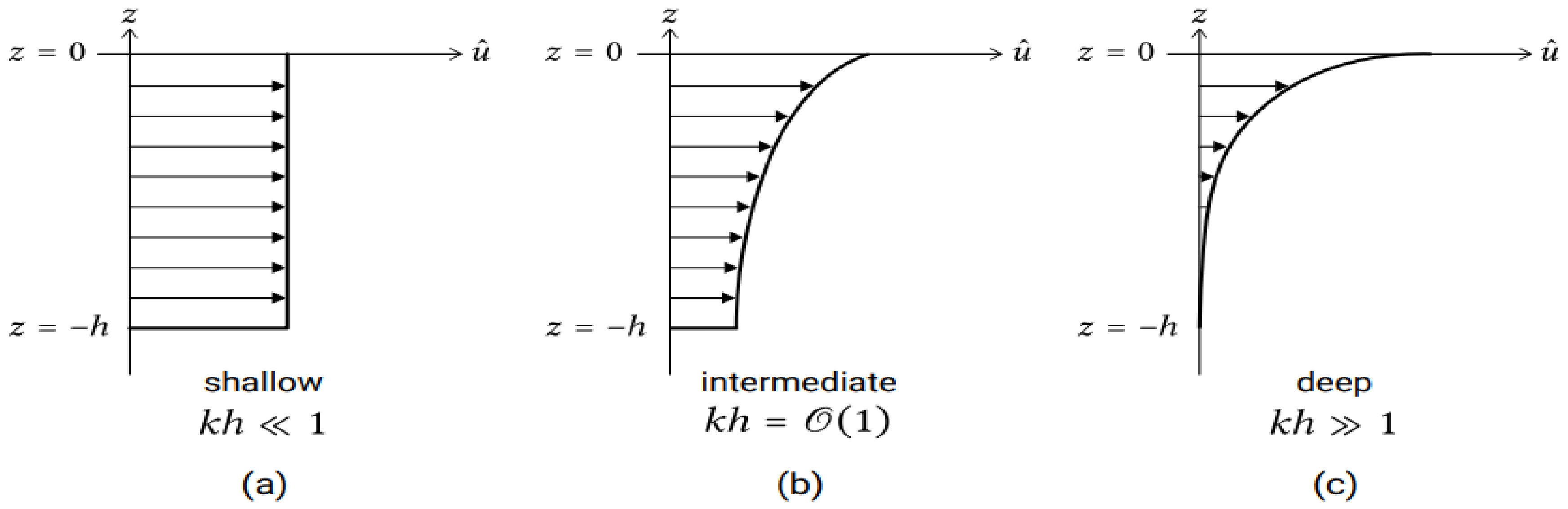
$\boldsymbol{L}$ = wavelength m

- The z-axis is defined positive upward with $z = 0$ at the surface and $z = -h$ at the bottom.

- The velocity u is in the wave propagation direction. Using the $\cosh kh$ and $\sinh kh$ approximations for shallow and deep water , the horizontal velocity profiles can be drawn schematically as in Figure.

- In shallow water ($kh \ll 1$; in practice $kh < \pi/10$ or $h/L > 1/20$), the depth-uniform velocity amplitude is given by :

$$\hat{\mathbf{u}} = \frac{\boldsymbol{\omega}\mathbf{a}}{\mathbf{kh}} = \mathbf{c}\frac{\mathbf{a}}{\mathbf{h}} = \sqrt{\mathbf{gh}}\frac{\mathbf{H}}{\mathbf{2h}}$$



- The **particle excursions** (i.e. the **horizontal** and **vertical displacements** of the particles) are the **time integrals** of the **oscillatory horizontal** and **vertical flow velocities** respectively.

- This means that the **amplitude** of the **horizontal particle excursion** is given by:

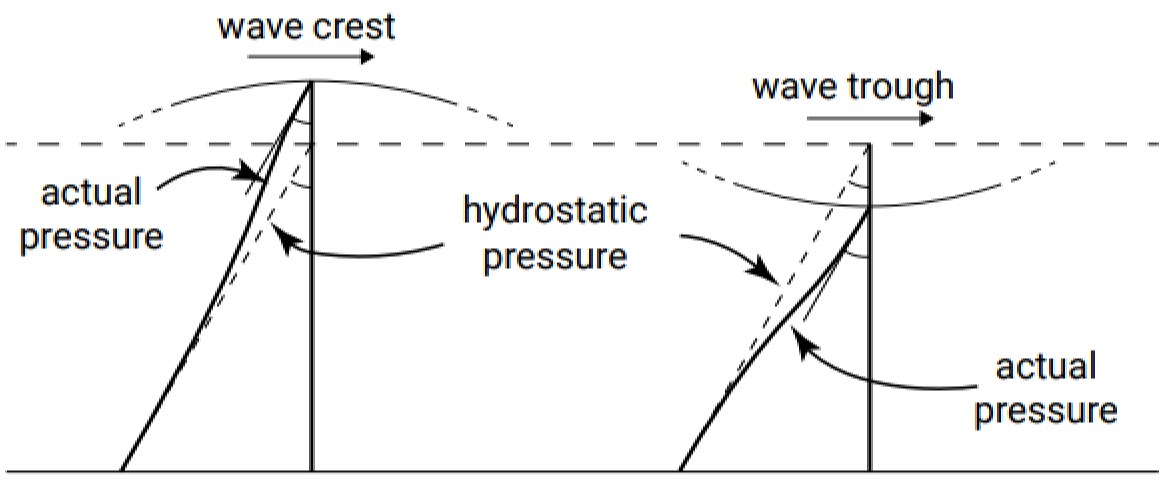
$$\hat{\boldsymbol{\xi}} = \frac{\hat{\boldsymbol{u}}}{\boldsymbol{\omega}}$$

Dynamic Pressure

- **Pressure gradients** in **coastal waters** are mainly due to **mean water level variations** (**hydrostatic pressure**) and **fluctuations of pressure due to waves**.

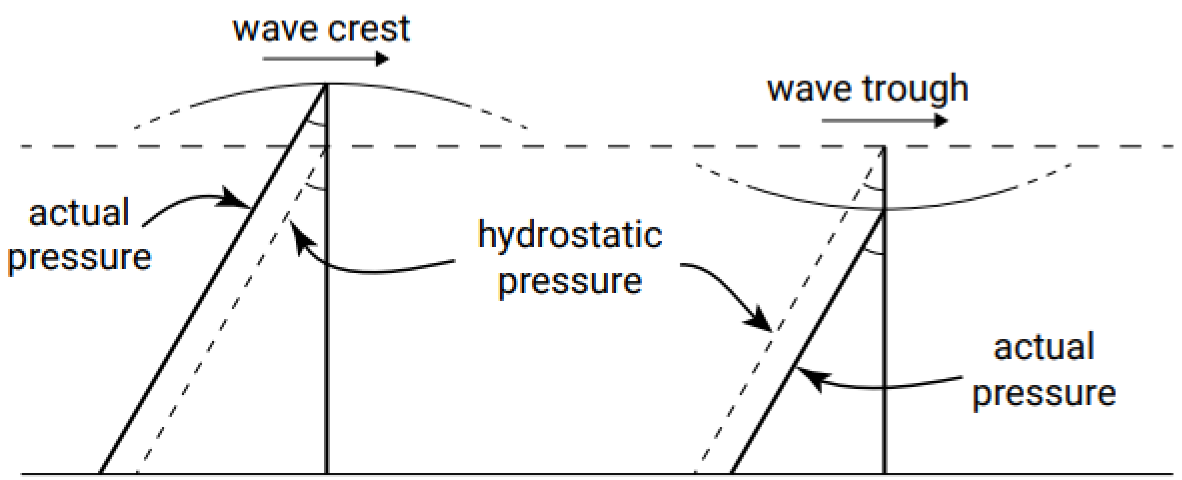
- The **hydrostatic pressure** due to **mean water level variation** is $\boldsymbol{p_0 = -\rho g z}$ and hence **linearly increases** from **zero** at the water surface $\boldsymbol{z = 0}$ **to** $\boldsymbol{p_0 = \rho g h}$ at the bottom $\boldsymbol{z = -h}$.

- In the case of waves, the **total pressure** is the **sum of this hydrostatic pressure** $\boldsymbol{p_0}$ (from $\boldsymbol{z = -h}$ to $\boldsymbol{z = \eta}$) **plus** the **wave-induced or dynamic pressure** $\boldsymbol{p_{\text{wave}}}$ from (from $z = -h$ to $z = \eta$).



(a) deep water

- **Wave-induced pressure oscillations** are **different** under **wave crest** and **wave trough** and – in **intermediate** and **deep water** – reduce with **depth** below the **free surface**.



(b) shallow water

- According to **linear theory**, the **wave-induced pressure** varies harmonically (in phase with the surface elevation η) with amplitude:

$$\hat{p} = \frac{\rho g H}{2}\frac{\cosh k(h+z)}{\cosh kh}$$

- **wave-induced pressure** which reduces in **shallow water** to:

$$\hat{p} = \frac{\rho g H}{2}$$

- Hence, in **shallow water** the **hydrostatic dynamic pressure** varies linearly with the **free surface elevation.**

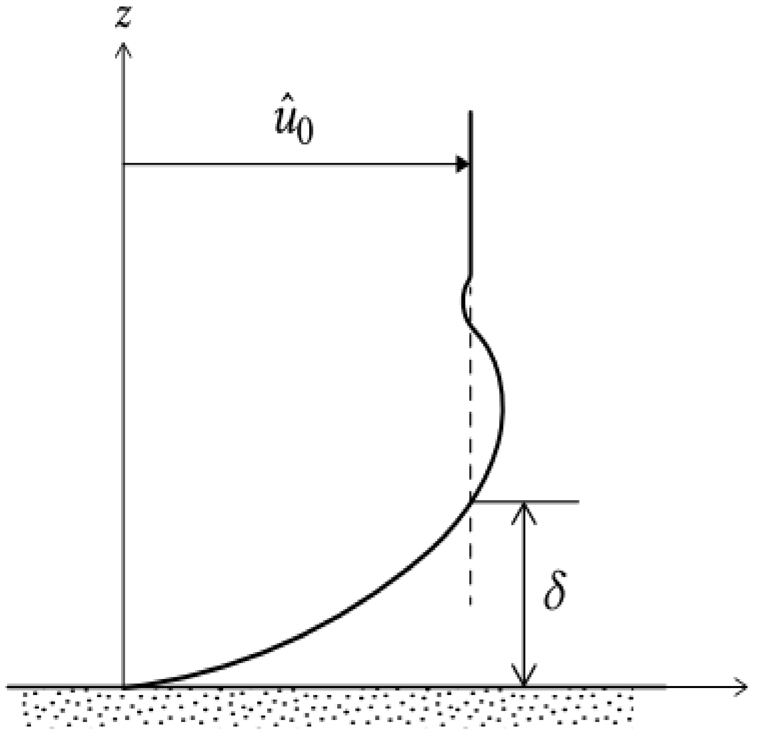
$$p_{\text{wave}} = \tilde{p} = \rho g \eta$$

- The **tilde** indicates the **purely oscillatory character**.
- To derive **wave-induced pressure** the **amplitude** was assumed to be **very small** in order to **linearize** the **free surface boundary condition**.
- It is therefore **not valid** in the **region** between the **trough** and the **crest elevation**.
- We can, however, assume that the **dynamic pressure** is **hydrostatic** between **wave trough** and **wave crest**:

$$\tilde{p} = \rho g \eta$$

Wave Boundary Layer

- Most wave theories are valid from the water level to a small distance from the bed, where the flow still is unaffected by the boundary.
- Closer to the bed, in a thin layer called the wave boundary layer(δ).
- vorticity (rotation) can be generated, which is not included in linear wave theory or in most other (irrotational) wave theories for that matter.
- The distance denoted as δ in Figure is the thickness of the wave boundary layer, the transition layer between the bed and the layer of ‘normal’ oscillating flow.



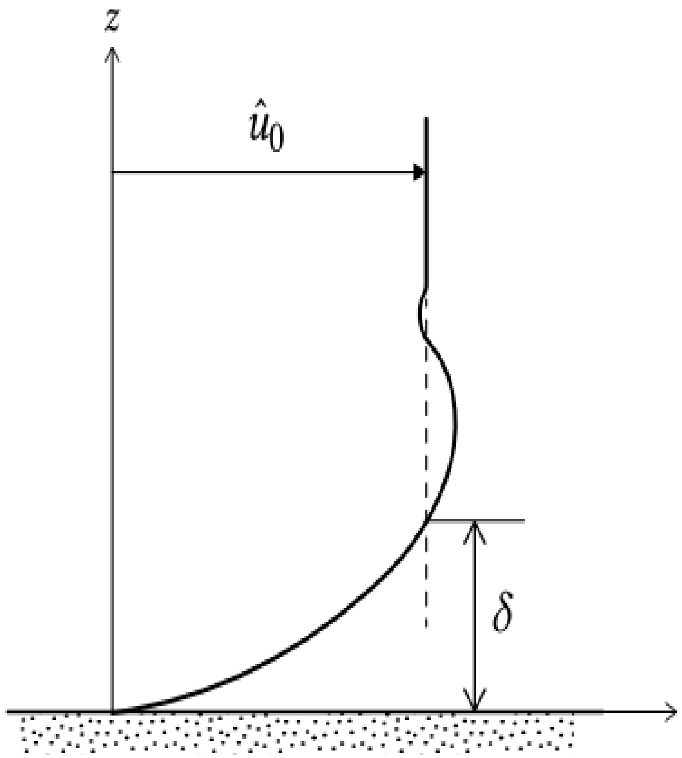
- The thickness is generally between 1 cm and 10 cm for short-period waves (T < 10 s).

- The reason for this small thickness is that there is not sufficient time for the layer to grow out in the vertical direction, because the current regularly reverses.

- It is typical for oscillating boundary layers that the maximum flow velocity near the bed is somewhat larger than the so-called free stream velocity.

- The free-stream velocity amplitude $\hat{u}_0$ according to linear theory for $z = -h$

$$\hat{u}_0 = \frac{\omega a}{\sinh kh}$$



- The flow in the wave boundary layer is generally turbulent due to the presence of roughness elements on the bed.
- The water moving along the bed incurs a shear stress on the bed.
- This becomes clear when imagining that – as a result of viscosity and turbulence – the flow sticks to the wall (no-slip condition).
- Hence, the orbital velocity increases from zero at the bed to the undisturbed free-stream velocity at the top of the wave boundary layer $z = \delta$.
- Because of the thin boundary layer, the velocity gradients perpendicular to the bed are large and give rise to large stresses in the wave boundary layer.
- The friction in the wave boundary layer results in dissipation of wave energy.

- In coastal waters, **turbulent stresses** – arising from **turbulent fluctuations** of the **velocity** are **much larger** than **viscous stresses** arising from **small-scale erratic movements of molecules**.

- Now think of the total **horizontal velocity u** and **vertical velocity w** to be composed of **a mean**, **a wave** and **a turbulent part**, hence

$$u = U + \tilde{u} + u'$$

$$w = W + \tilde{w} + w'$$

- **Turbulent shear stress** is defined as the **stress introduced** when **averaging over the turbulent motion**:

$$\tau(z) = \overline{\rho u' w'}$$

For practical purposes, the following aspects of the wave boundary layer is to be consider:

- The water moving along the bed incurs a shear stress on the bed.
- The orbital motion under waves, even without the presence of a uniform current, gives a time-varying shear stress at the bed, which can set sediment grains into motion.
- Due to bed friction, the wave boundary layer dissipates energy from the flow above .
- The wave-induced streaming should be taken into account for net sediment transport computations.

Bed shear stress

- **Bed shear stress** (τ) is the stress exerted by the water flow on the bed of a water body, which can vary due to the presence of waves and currents.
- It is crucial for understanding **sediment transport**.
- If we refrain from **turbulence modelling**, (bed) **friction** is a **major unknown** that has to be **determined** using (**empirical**) **friction laws**.
- This introduces **coefficients** that need to be **calibrated,** which makes the calibration of these models important.
- To determine the **bed shear stress**, **Jonsson** (**1967**) introduced the **concept** of a **wave friction factor** in analogy with the **current friction factor**.

- The **current friction factor** relates the **bed shear stress** to the **depth-averaged current velocity**, whereas the **wave friction factor** relates the **bed shear stress** to the **free stream velocity**.
- For a **current only**, the *magnitude* of the **bed shear stress** is $\boldsymbol{\tau_c = c_f \rho U^2}$
- The **friction factor** $\boldsymbol{c_f}$ is a dimensionless coefficient relating the bed shear stress to the square of the velocity.
- $\boldsymbol{c_f}$ can be expected to **depend** on the **bed material** and **bed forms** (**bed roughness**)
- For **uniform flow** in a **canal** driven by a **small slope** of the **mean water surface**, Chezy's derived theoretically a **Chezy's coefficient**

$$\boldsymbol{C = 18 \log 12 \frac{h}{r}},$$

where

$\boldsymbol{r}$ is the **bottom roughness** and
$\boldsymbol{h}$ is the **water depth**.

- The **Chezy's coefficient** C relates to c_f as follows:

$$c_f = \frac{g}{C^2}$$

- Under **waves**, the **bed shear stress** varies in **time** and **reverses** with the **direction** of the **orbital velocities**.

- For **linear waves** with a **free stream velocity**

$$u = \widehat{u_0} \cos \omega t$$

- Jonsson defined the **friction factor** f_w through the following formula for the **magnitude** of the ***maximum*** **bed shear stress**:

$$\hat{\tau}_w = 0.5 \rho f_w \hat{u}_0^2$$

- For a **rough bed** and **turbulent flow**, it is **not easy** to determine the **friction coefficient** as it **cannot** be **measured directly**.

- The **friction coefficient** will generally **depend** on the **bed material** and the **bed forms** (e.g. ripples).

- The following **variables** can be found in expressions for $\boldsymbol{f_w}$ for **rough turbulent flow**.

- the **bed roughness** $\boldsymbol{k_s}$ (**Nikuradse roughness**) or $\boldsymbol{r}$ of **the wall**; the bed roughness represents the size of the roughness elements, for instance the grains.

- the **particle excursion amplitude** close to the **bed** $\boldsymbol{\hat{\xi}_0 = \widehat{u_0} / \omega}$.

- Let us consider **two cases** in which $\widehat{u_0}$ is the **same**, but T **differs**.
- For the case with a **large** value for T , the **value** of $\hat{\xi_0}$ is **larger** than in the case with the **lower value** for T .
- Assuming an **equal value** for **r**, this **means** that the case with the **larger value** for the **wave period** gives **lower values** for the **wave friction factor**.
- This can be understood by considering that the **boundary layer thickness varies with time**.
- In case of a **larger wave period**, more time is available to **develop the boundary layer**, which therefore reaches a **larger maximum thickness**.
- Consequently, the **velocity gradients** in the **boundary layer** are **smaller**, leading to a **smaller maximum shear stress** and **friction factor**.

- A frequently applied formula for f_w is that of **Jonsson** (1967), **rewritten** by **Swart** (1974) into:

$$f_w = exp\left[-5.977 + 5.213\left(\frac{\hat{\xi}_0}{r}\right)^{-0.194}\right]$$

$$f_w = 0.30 \text{ for } \left(\frac{\hat{\xi}_0}{r}\right) < 1.59$$

- Many **simpler formulas** exist, such as (**Soulsby, 1994**):

$$f_w = 1.39\left(\frac{\hat{\xi}_0}{r/30}\right)^{-0.52}$$

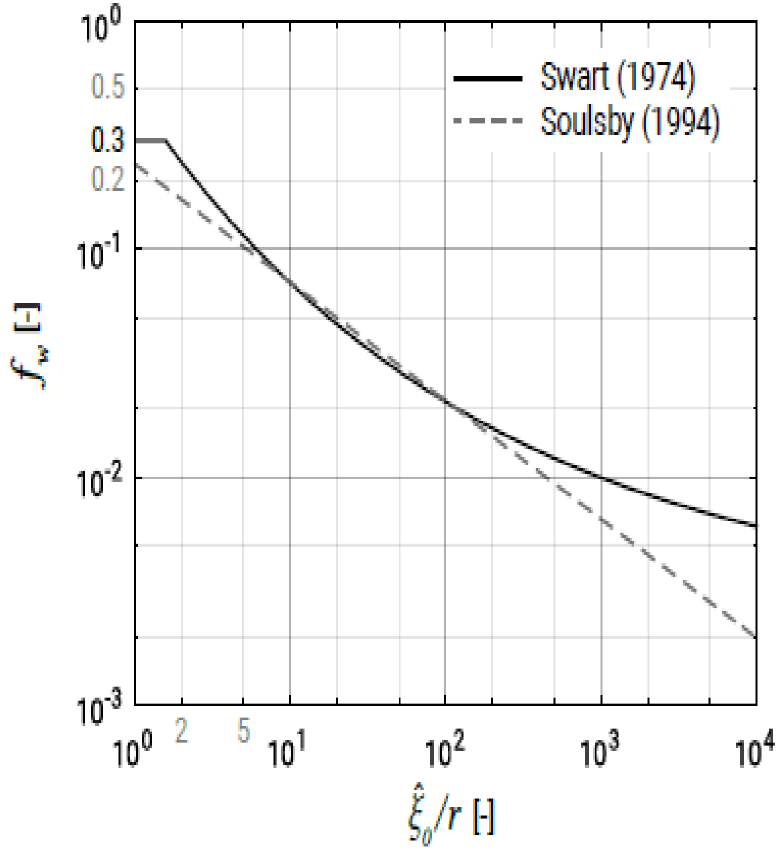
$$f_{w,max} = 0.3$$

- For the **roughness value** r the so-called **Nikuradse roughness** k_s is often used, which is normally set as a function of the grain size.

- As expected, the **friction factor f_w increases** if $\hat{\xi}_0/r$ **decreases**.

- The **upper limit** of f_w has been questioned by various researchers: some suggest that there is no **upper limit** and that the **friction factor** remains **proportional** to $\hat{\xi}_0/r$.

- In case of **irregular waves**, the near-bed orbital velocity amplitude in the above formulas should be based on the **root-mean-square wave height and the wave orbital** excursion parameter near the bed on the root-mean-square wave height and peak period.



Problem

Find the maximum shear stress near the bed under waves only from the linear wave theory for given parameters.

(Water depth d = 3 m, Roughness height r = 0.06 m, Wave height H = 1.18 m, Wave period T = 8 s)

Solution : The amplitude of the velocity near the bed can be found from linear wave theory:

$$\hat{u}_0 = \frac{\omega H}{2}\frac{1}{\sinh kd} \text{ and } \hat{\xi}_0 = \frac{\hat{u}_0 T}{2\pi}$$

$$L_o = 1.56T^2 = 1.56 \times 8^2 = 99.84 \text{ m}$$

$$\frac{d}{L_0} = \frac{3}{99.84} = 0.03$$

$$\frac{d}{L} = 0.0713 \text{ (from wave table)}$$

$$\frac{3}{L} = 0.0713 \quad \text{or, L = 42.08 m}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{8} = 0.7$$

$$\text{k} = \frac{2\pi}{L} = \frac{2\pi}{42.08} = 0.149$$

d/L_o	d/L	$2\pi d/L$	tanh ($2\pi d/L$)	sinh ($2\pi d/L$)	cosh ($2\pi d/L$)	H/H'_o	K	$4\pi d/L$	sinh ($4\pi d/L$)	cosh ($4\pi d/L$)	n	C_G/C_o	M
0.0110	0.0423	0.2660	0.2599	0.2691	1.0356	1.4032	0.9656	0.5319	0.5574	1.1448	0.9772	0.2539	73.08
0.0120	0.0443	0.2781	0.2711	0.2817	1.0389	1.3752	0.9625	0.5562	0.5853	1.1587	0.9751	0.2644	67.13
0.0130	0.0461	0.2897	0.2819	0.2938	1.0423	1.3501	0.9594	0.5795	0.6125	1.1727	0.9731	0.2743	62.10
0.0140	0.0479	0.3010	0.2922	0.3056	1.0457	1.3274	0.9563	0.6020	0.6390	1.1868	0.9710	0.2838	57.78
0.0150	0.0496	0.3119	0.3022	0.3170	1.0490	1.3068	0.9533	0.6238	0.6651	1.2010	0.9690	0.2928	54.05
0.0160	0.0513	0.3225	0.3117	0.3281	1.0525	1.2879	0.9502	0.6450	0.6906	1.2153	0.9669	0.3014	50.78
0.0170	0.0530	0.3328	0.3210	0.3389	1.0559	1.2706	0.9471	0.6655	0.7157	1.2298	0.9649	0.3097	47.89
0.0180	0.0546	0.3428	0.3300	0.3495	1.0593	1.2545	0.9440	0.6855	0.7405	1.2443	0.9629	0.3177	45.33
0.0190	0.0561	0.3525	0.3386	0.3599	1.0628	1.2396	0.9409	0.7051	0.7650	1.2590	0.9609	0.3254	43.04
0.0200	0.0576	0.3621	0.3471	0.3701	1.0663	1.2258	0.9378	0.7242	0.7892	1.2739	0.9588	0.3328	40.97
0.0210	0.0591	0.3714	0.3552	0.3800	1.0698	1.2129	0.9348	0.7429	0.8131	1.2888	0.9568	0.3399	39.10
0.0220	0.0606	0.3806	0.3632	0.3898	1.0733	1.2008	0.9317	0.7612	0.8368	1.3039	0.9548	0.3468	37.41
0.0230	0.0620	0.3896	0.3710	0.3995	1.0768	1.1894	0.9286	0.7791	0.8603	1.3192	0.9528	0.3535	35.86
0.0240	0.0634	0.3984	0.3785	0.4090	1.0804	1.1787	0.9256	0.7967	0.8837	1.3345	0.9508	0.3599	34.44
0.0250	0.0648	0.4070	0.3859	0.4183	1.0840	1.1686	0.9225	0.8140	0.9070	1.3500	0.9488	0.3662	33.13
0.0260	0.0661	0.4155	0.3932	0.4276	1.0876	1.1590	0.9195	0.8310	0.9301	1.3657	0.9468	0.3722	31.93
0.0270	0.0675	0.4239	0.4002	0.4367	1.0912	1.1500	0.9164	0.8478	0.9531	1.3814	0.9448	0.3781	30.81
0.0280	0.0688	0.4321	0.4071	0.4457	1.0948	1.1414	0.9134	0.8643	0.9760	1.3973	0.9428	0.3838	29.77
0.0290	0.0701	0.4403	0.4139	0.4546	1.0985	1.1332	0.9103	0.8805	0.9988	1.4134	0.9408	0.3894	28.81
0.0300	0.0713	0.4483	0.4205	0.4634	1.1022	1.1254	0.9073	0.8966	1.0216	1.4296	0.9388	0.3948	27.91

$$\hat{u}_0 = \frac{\omega H}{2}\frac{1}{\sinh kd} = \frac{0.786 \times 1.118}{2} \times \frac{1}{\sinh(0.149 \times 3)} = 1.003m/s$$

$$\hat{\xi}_0 = \frac{\hat{u}_0 T}{2\pi} = \frac{1.003 \times 8}{2\pi} = 1.27m$$

For a value of $\hat{\xi}_0 /r$ > 1.59 the friction factor equals:

$$f_w = \exp\left[-5.977 + 5.213\left(\hat{\xi}_0/r\right)^{-0.194}\right] = 0.045$$

The maximum bottom shear stress follows from:

$$\hat{\tau}_w = \frac{1}{2}\rho f_w \hat{u}_0^2 = 22.5\mathrm{N/m^2}$$

Hence, for a maximum near-bed orbital velocity of 1 m/s, the bottom shear stress due to waves equals 22.5 N/m^2.

Now consider the situation of a mean current of 1 m/s.

Assume that the roughness height and water depth are the same as above.

Using $C = 18 \log 12d/r = 50\ \mathrm{m}^{1/2}/\mathrm{s}$

we find:

$$\tau_c = \rho \frac{g}{C^2} U^2 = 3.9\mathrm{N/m}^2$$